

Designing a Human-Wearable Moulage for Hemorrhage Control Wound Packing Training

Uncontrolled hemorrhage is the leading cause of preventable death on the battlefield. Currently there are advanced training methods for tourniquet use; however, in junctional wounds such as at the neck and groin, tourniquets cannot be used. As a result, wound packing must be performed, but current training relies on static manikins that lack real-time feedback, limiting trainees' ability to assess proper pressure and technique. This project aims to address this gap through the design of a human-wearable moulage system that simulates a bleeding gunshot wound and provides real-time performance feedback.

The device integrates a silicone wound model, a peristaltic pump to simulate arterial bleeding, embedded force sensors to measure applied pressure, and an Arduino-based system with an LCD interface. A control algorithm links pressure to simulated blood flow: adequate pressure triggers an exponential decrease in flow, while insufficient pressure causes flow to increase linearly back to the initial flow rate with the LCD screen displaying whether sufficient pressure is being applied or not, providing real-time feedback and reinforcing correct wound packing technique. Testing demonstrated reliable sensor calibration and accurate flow control. Trials showed that sustained adequate pressure effectively reduced simulated hemorrhage, while inadequate pressure led to increased flow, confirming system responsiveness.

This work demonstrates a novel training tool that combines wearability, physiological simulation, and real-time feedback, offering a significant improvement over existing methods. By providing objective performance metrics and realistic practice conditions, this system has the potential to enhance hemorrhage control training and improve outcomes in emergency care on the battlefield.